Explainable and Data-Efficient Deep Learning for Enhanced Attack Detection in IIoT Ecosystem

Danish Attique, Wang Hao, Wang Ping, Danish Javeed, and Prabhat Kumar, *Member, IEEE*

***Abstract*—The Industrial Internet of Things (IIoT) is rapidly evolving, and with this evolution, cyber threats have become a significant issue. IIoT networks, despite improving service quality, are uniquely vulnerable to security threats due to their inherent connectivity and the use of low-power devices. Traditional Deep Learning-based intrusion detection system (IDS), while accurate, suffer from a "black box" issue that hides the reasoning behind their decisions, leading to a decrease in user trust. To address this, our research presents an Explainable and intelligent mechanism for data-efficient intrusion detection in IIoT. Our proposed IDS enhances data efficiency by employing a bidirectional long-short term memory (BiLSTM) model with a self-adaptive attention mechanism. The self-adaptive attention mechanism is a novel feature of our IDS framework, designed specifically for IIoT environments. This mechanism dynamically adjusts its focus to prioritize critical elements within a dataset, allocating more computational resources to data segments likely to contain patterns or anomalies indicative of security threats. When integrated with BiLSTM, which excels at capturing temporal dependencies, the mechanism enhances the IDS's ability to learn efficiently from limited datasets. This focus on significant data features and temporal patterns reduces the need for extensive training datasets, making it particularly effective in IIoT settings where data may be sparse yet complex. In addition, we enhance the proposed IDS's transparency by incorporating the Shapley additive explanations mechanism from explainable AI, thereby boosting the IDS's trustworthiness and interpretability. Our system exhibits outstanding performance on benchmark datasets, such as CICIDS2017 and X-IIoTID, attaining accuracies of 99.92% and 96.54%, respectively.**

***Index Terms*—Cyber attacks, explainable AI (XAI), intrusion detection system (IDS), Industrial Internet of Things (IIoT), proactive defense.**

I. Introduction

THE INDUSTRIAL Internet of Things (IIoT) offers the potential for enhancing service quality for new applications because of the rise in the data generated by its connected devices [1]. IIoT and Industry 4.0 are concepts that might be closely connected. Intending to boost industries, this idea is quickly introducing new trends in expanding industrial concepts, logistical services, industrial processes, and automation [2]. The IIoT network comprises thousands of devices and communication protocols [3]. Communication sensors are a crucial element responsible for collecting, processing, and transmitting data in an IoT-based industrial environment. An abundance of parameters is responsible for efficient IoT communication in smart industries, i.e., robustness, minimal latency, and communication integrity. These parameters can be addressed by appropriate load balancing and scheduled communication among the participant nodes [4]. Modern industries can gather and analyze enormous amounts of data to come to intelligent business judgments due to a variety of devices, sensors, actuators, and controllers. These sensors and devices have limited memory, power, and communication resources, hence they are considered resource-constrained. IIoT has been used by several industries in recent years, including manufacturing, healthcare, agriculture, and unmanned aerial vehicles [5]. However, the prevalent nature of increased connected devices and the resource-constrained nature of IIoT makes them vulnerable to cyber attacks [6]. One compromised device may communicate malicious data to cloud servers or permit unauthorized access to confidential company data, such as business plans and trade information, resulting in data loss and resource corruption. Such an incident may lead the industry to financial loss, operational disruption, and reputation loss [7].

Ensuring IIoT security is one of the most important problems in modern industrial operations. It encompasses safeguarding communication and physical privacy, protecting edge devices from viruses, and preventing unwanted access. The IIoT's security, privacy, and dependability may be increased by implementing sophisticated and strong security frameworks [8]. Intrusion detection system (IDS) is one of the most often utilized security and privacy technologies in these sectors. It searches for particular signatures and patterns in network traffic to detect malicious activity and unauthorized access [9]. However, such an IDS cannot detect sophisticated and advanced persistent threats (APTs) [10]. Furthermore, the resource-constrained nature of devices in the IIoT environment makes it more challenging

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to design a lightweight and efficient IDS for its security. Different researchers have employed DL and ML to design an automated, lightweight, and resilient IDS for IIoT in the past few years. IDS powered by DL has proven itself to be efficient by achieving high-detection accuracy [11]. However, such IDS are considered black boxes by security analysts and users due to their complex nature and lack of explanation and interpretation for their decisions, such as gaining an understanding of the primary data proof of the decision-making processes for the proposed model behavior [12]. As a result, there is a lack of trust in the model's decision, and their output cannot be utilized to further enhance the behavior and logic provided by the model.

Explainable AI (XAI) is grabbing significant consideration because of its advanced attributes that overshine conventional ML/DL concepts [13]. XAI-based IDS provides consumers with systematic and understandable reasons for its actions. It has become crucial for interpreting the decisions of ML/DL-based models by enabling security specialists and developers to comprehend the primary data and the influence of the malevolent data in any IDS. The current literature heavily emphasizes the detection accuracy of DL-based IDS. However, they do not provide the reasoning and explanations of the decisions provided by their proposed IDS [14]. Hence, their model lacks trust and transparency. Additionally, the XAI-based IDS is still in its early stages and needs improvement. Motivated by this, we propose an explainable DL-based IDS for enhanced trust management and interpretation by providing the interpretation of the IDS decision-making process to help the security analysts understand the rationale behind the decision of our proposed IDS.

A. Contribution

The aim of this work is to develop a proactive threat detection framework for IIoT networks to maintain their security and privacy. This study presents an AI-based Intelligent threat detection mechanism to identify and prevent hidden threats in such a network. The following are the contributions of this work.

1) The research presents a novel IDS specifically designed for the IIoT. The proposed IDS enhances data efficiency using a bidirectional long-short term memory (BiLSTM) model integrated with a self-adaptive attention mechanism. This mechanism dynamically prioritizes critical elements within a dataset, allocating more computational resources to data segments likely to contain patterns or anomalies indicative of security threats.
2) We further employed the SHAP mechanism of the XAI to explain and interpret the decision of the proposed IDS. Such an explanation and interpretation will aid the users in understanding the rationale behind the IDS decision and aid the security operation center (SOC) analysts in designing new and efficient IDS for the industrial networks.
3) We evaluate the proposed IDS's performance using the CICIDS2017 and X-IIoTID datasets and compare its performance with some traditional and recent frameworks from the literature. The performance comparison endorses the dominance of the proposed IDS against these frameworks and proves it an efficient IDS for such industries.

This work is further organized as follows. Section II discusses the relevant literature and its limitations. Section III presents the details of the proposed IDS for IIoT. Section IV presents the experimental setup. Section V analyze the experimental results, performance analysis, and comparison of the proposed IDS. Finally, in Section VI, we conclude this research work.

II. Existing Literature

Intrusion detection in IoT-empowered smart Industries using DL is a concept that is attracting considerable attention in the present era. The literature is surrounded by abundant such studies, and we have addressed some of them here. Wei et al. [15] employed a hybrid DL autoencoder combined with multilayer perceptron (MLP). The proposed model is capable of extracting the features automatically. They trained their proposed framework using the publicly available CICDDoS2019 dataset and attained 98.34% detection accuracy. Yungaicela-Naula et al. [27] used a DL-based model for implementing a modular software-defined networking (SDN)-based network to provide a solution for DDoS detection in transport and application layers. Li et al. [16] recommended a Multi-CNN fusion-based scheme for efficient attack detection in IoT networks. They trained and evaluated their proposed technique using the NSL-KDD dataset, where the model recognized the attack classes successfully. They compared the performance of their proposed model with ML-based techniques and claimed to outclass them with 86.95% accuracy.

De Souza et al. [17] suggested an IDS for the IIoT environments. They employed a hybrid model of DNN and K-nearest neighbors (KNNs) for binary classification. The authors used the NSL-KDD and CICIDS datasets for experimentation purposes. Alkahtani and Aldhyani [18] proposed a hybrid model based on CNN and LSTM. They used the real traffic of nine commercial IoT devices, infected with Bashlite and Mirai botnet attacks. Their proposed model uses word embedding to identify and change text representation to an integer. The model shows efficiency in threat detection. However, it has a high-processing time and resource overhead. Pooja and Shrinivasacharya [26] employed a Bidirectional LSTM-based model for efficient threat detection. They employed the KDDCUP-99 and UNSW-NB15 datasets to train and evaluate their proposed model. Further, Jithu et al. [19] proposed an intelligent IDS based on DNN and employed RELU as an activation function to detect and mitigate DDoS attacks in the industrial network. Their model accomplished an accuracy of 94%. Li et al. [20] employed a federated learning (FL)-based IDS combined with Edge computing to identify botnets in the IIoT network and attain an efficient detection accuracy of 98%. Furthermore, Oseni et al. [21] designed an IDS for IoT-enabled large-scale vehicular networks. The proposed framework incorporates Shapley additive explanations (SHAPs) that provide comprehensive insights regarding

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TABLE I
LITERATURE OVERVIEW

Reference	Approach	Model Used	Dataset	Multiclass Classification	Explainable AI (XAI)	Limitations
[15]	DL	Autoencoder, MLP	CICDDoS2019	✓	×	The proposed model is black-box
[16]	DL	Multi-CNN	NSL-KDD	✓	×	Obsolete dataset
[17]	DL/ML	DNN, KNN	NSL-KDD, CICIDS2017	×	×	The model is black-box and lacks comparison
[18]	DL	CNN, LSTM	N-BaIoT	✓	×	Not specified
[19]	DL	DNN	BoT-IoT	✓	×	Comparison is not provided
[20]	FL	-	UNSWNB-15	×	×	Black-box model
[21]	DL	CNN	ToN_IoT	✓	✓	Computationally expensive and intensive
[22]	ML/DL	XGBoost, LSTM, GRU	MQTT	✓	×	No comparison
[23]	DL	FF, MC	KDD-Cup-1999, UNSW-NB15, WSN-DS, CICIDS-2017	✓	×	Computationally complex and black-box
[24]	DL	MLP	WUSTL-IIOT-2021	×	✓	Not specified
[25]	DL	CNN	BoT-IoT, IoT Network Intrusion, MQTT-IoT-IDS2020, IoT-23	×	×	The model is black-box
[26]	DL	BiLSTM	KDD-CUP99, UNSW-NB15	✓	×	Black-box model

Terms & Abbreviations: MLP: Multilayer perception; CNN: Convolutional Neural Network; KNN: k-nearest Neighbors; LSTM: Long short-term Memory Networks; FF: Feature Fusion; MC: Meta Classifier.

the procedural flow of DL. The authors trained their model on the ToN-IoT dataset that contains detailed impressions of frequently occurring attacks in the respective Internet of Vehicles (IoV) environment. The performance regarding threat detection accuracy and other evaluation metrics are examined. Additionally, the performance is compared with some phenomenal threat detection frameworks from the literature. The proposed model seems to outclass existing intrusion detection schemes.

Alaiz-Moreton et al. [22] used LSTM-based IDS to detect DoS and MiTM attacks in IoT systems and attained an accuracy of 92.76%. Likewise, Ravi et al. [23] developed an intelligent IDS based on the DL feature fusion (FF) approach combined with an ensemble meta classifier (MC) and attained 99% accuracy, respectively. Alani et al. [24] employed XAI and DL to develop an efficient and robust IDS (DeepIIoT) for IIoT networks. They employed the WUSTL-IIoT2021 dataset to train and evaluate their proposed model and achieved an enhanced detection accuracy of 99%.

Security analysis of DDoS attacks in IoT (SAD-IoT) is a searching framework designed by aggregating various DL algorithms. The framework is evaluated and trained on BoT-IoT and UNSW datasets where the system achieved a noticeable threat detection accuracy [28]. An intelligent attack detection scheme to provide extensive security in edge IIoT devices is presented in [29]. The designed scheme used the best characteristics from spider monkey (SM), principal component analysis (PCA), and correlation attribute evaluation (CAE) to analyze traffic patterns. The model demonstrates noticeable accuracy and precision in terms of efficient attack detection. In [25], another anomaly detection mechanism is presented to safeguard IIoT-based networks from advanced threats. The authors used the MQT-IoT-IDS2020 dataset to train and evaluate their CNN-based module. Upon evaluation, it achieved an attack detection accuracy of 98.80%. A variety of DL and ML-based IDS has been proposed by researchers in the past few years as summarized in Table I.

III. PROPOSED EXPLAINABLE-DEEP LEARNING (XDL)-BASED IDS FOR IIOT

In this section, we present the methodology of the proposed IDS. We first describe BiLSTM, followed by the self-adaptive attention Mechanism. Further, we provide details about the XAI mechanism. Finally, we provide complete details of hyperparameters in this section. The proposed scheme automatically learns and upholds the vital features of flow-based network traffic from the CICIDS2017 and X-IIoTID datasets to detect intrusions and cyber-attacks. Further, the proposed IDS assists security analysts by relieving them of strenuous investigation tasks and protecting the safety of such enterprises from cyber-attacks, ensuring their safety against cyber threats and intrusions. Algorithm 1 provides the proposed IDS's working mechanism.

A. BiLSTM

It refers to a type of RNN architecture that incorporates both forward and backward information flow. In a basic LSTM model, information flows in only one direction, either from past to future or from future to past. In a simple LSTM model, the flow of information is in one direction, i.e., from past to future or future to past. However, in BiLSTM, the input sequence (*IS*) is processed in both directions at the same time, allowing the model to capture dependencies from both the past and the future. BiLSTM comprises two layers of LSTM, one processes the *IS* in forward ($\rightarrow$), and the other processes it in the backward direction ($\leftarrow$). The forward layer receives the *IS* and computes its hidden state ($\mathbb{h}_t$) and output by considering the past context. On the other hand, the backward layer receives the *IS* in reverse order and computes its output by considering the future context. The information flow in both directions makes the BiLSTM effective in capturing the long-range context and dependencies in the IS. It uses its gating mechanism, i.e., forget gate (For_{Gate}), input gate (Inp_{Gate}), and output gate (Out_{Gate}) to solve the problem of exploding gradients. The following equations perform the operations of the BiLSTM's forward process ($\rightarrow$) [30]:

$$\overrightarrow{\text{For}_{\text{Gate}}} = \alpha\left(\overrightarrow{\mathbb{W}_{\text{For}}} * \left[\overrightarrow{\mathbb{h}_{t-1}}, \overrightarrow{\mathbb{P}_{\text{For}}}\right] + \overrightarrow{\mathbb{B}_{\text{For}}}\right) \tag{1}$$

$$\overrightarrow{\text{Inp}_{\text{Gate}}} = \alpha\left(\overrightarrow{\mathbb{W}_{\text{Inp}}} * \left[\overrightarrow{\mathbb{h}_{t-1}}, \overrightarrow{\mathbb{P}_{\text{Inp}}}\right] + \overrightarrow{\mathbb{B}_{\text{Inp}}}\right) \tag{2}$$

$$\overrightarrow{C_Inp_t} = \tanh\left(\overrightarrow{\mathbb{W}_c} * \left[\overrightarrow{\mathbb{h}_{t-1}}, \overrightarrow{\mathbb{P}_c}\right] + \overrightarrow{\mathbb{B}_c}\right) \tag{3}$$

$$\overrightarrow{C_t} = \left(\overrightarrow{\text{For}_{\text{Gate}}} * \overrightarrow{c_{t-1}}\right) + \left(\overrightarrow{\text{Inp}_{\text{Gate}}} * \overrightarrow{C_Inp_t}\right) \tag{4}$$

$$\overrightarrow{\text{Out}_{\text{Gate}}} = \alpha\left(\overrightarrow{\mathbb{W}_{\text{Out}}} * \left[\overrightarrow{\mathbb{h}_{t-1}}, \overrightarrow{\mathbb{P}_{\text{Out}}}\right] + \overrightarrow{\mathbb{B}_{\text{Out}}}\right) \tag{5}$$

$$\overrightarrow{\mathbb{h}_t} = \overrightarrow{\text{Out}_{\text{Gate}}} * \tanh\left(\overrightarrow{C_t}\right) \tag{6}$$

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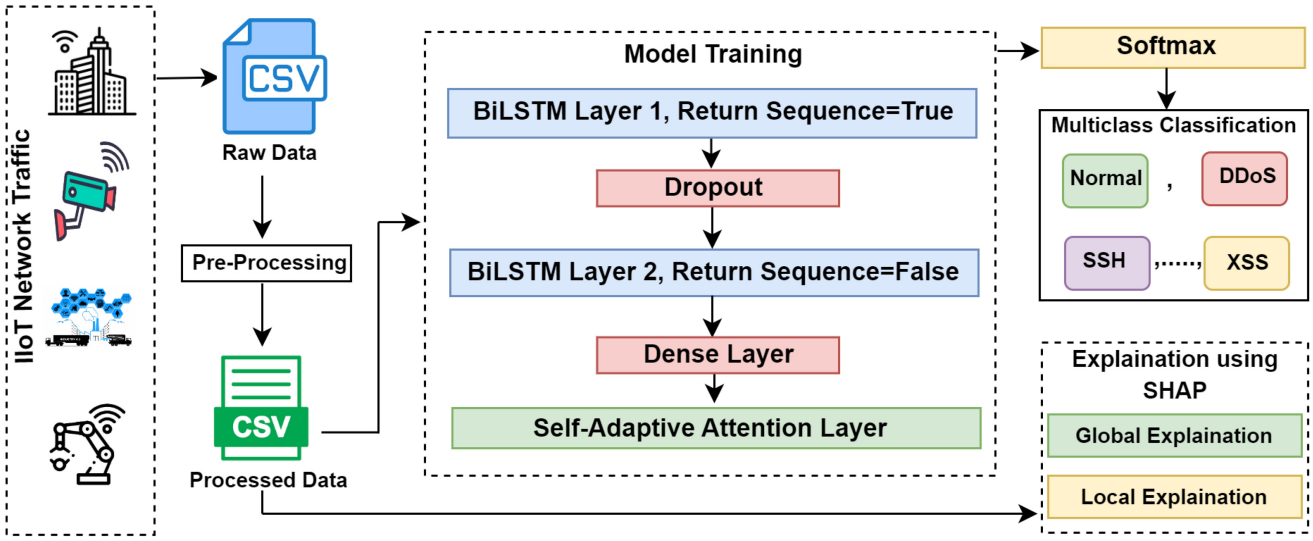


Fig. 1. Proposed IDS architecture.

where $\overrightarrow{C_t}$ is the cell state, $\overrightarrow{C_Inp_t}$ represents the candidate for $\overrightarrow{C_t}$ and $\overrightarrow{\mathbb{P}_{\text{For}}}$, $\overrightarrow{\mathbb{P}_{\text{Inp}}}$, $\overrightarrow{\mathbb{P}_c}$, and $\overrightarrow{\mathbb{P}_{\text{Out}}}$ denotes the inputs for the $\overrightarrow{\text{For}_{\text{Gate}}}$, $\overrightarrow{\text{Inp}_{\text{Gate}}}$, $\overrightarrow{C_Inp_t}$, and $\overrightarrow{\text{Out}_{\text{Gate}}}$, while $\overrightarrow{\mathbb{h}_{t-1}}$ represents the prior LSTM block's output for the ($\rightarrow$) process. The $\overrightarrow{\mathbb{W}_{\text{For}}}$, $\overrightarrow{\mathbb{W}_{\text{Inp}}}$, $\overrightarrow{\mathbb{W}_c}$, and $\overrightarrow{\mathbb{W}_{\text{Out}}}$ are the weights. Further, σ denotes the sigmoid activation function and $\overrightarrow{\mathbb{B}_{\text{For}}}$, $\overrightarrow{\mathbb{B}_{\text{Inp}}}$, $\overrightarrow{\mathbb{B}_c}$, and $\overrightarrow{\mathbb{B}_{\text{Out}}}$ are the Bias terms.

Furthermore, the following equations are used to perform the operations of the BiLSTM's backward process ($\leftarrow$):

$$\overleftarrow{\text{For}_{\text{Gate}}} = \alpha\left(\overleftarrow{\mathbb{W}_{\text{For}}} * \left[\overleftarrow{\mathbb{h}_{t+1}}, \overleftarrow{\mathbb{P}_{\text{For}}}\right] + \overleftarrow{\mathbb{B}_{\text{For}}}\right) \tag{7}$$

$$\overleftarrow{\text{Inp}_{\text{Gate}}} = \alpha\left(\overleftarrow{\mathbb{W}_{\text{Inp}}} * \left[\overleftarrow{\mathbb{h}_{t+1}}, \overleftarrow{\mathbb{P}_{\text{Inp}}}\right] + \overleftarrow{\mathbb{B}_{\text{Inp}}}\right) \tag{8}$$

$$\overleftarrow{C_Inp_t} = \tanh\left(\overleftarrow{\mathbb{W}_c} * \left[\overleftarrow{\mathbb{h}_{t+1}}, \overleftarrow{\mathbb{P}_c}\right] + \overleftarrow{\mathbb{B}_c}\right) \tag{9}$$

$$\overleftarrow{C_t} = \left(\overleftarrow{\text{For}_{\text{Gate}}} * \overleftarrow{c_{t+1}}\right) + \left(\overleftarrow{\text{Inp}_{\text{Gate}}} * \overleftarrow{C_Inp_t}\right) \tag{10}$$

$$\overleftarrow{\text{Out}_{\text{Gate}}} = \alpha\left(\overleftarrow{\mathbb{W}_{\text{Out}}} * \left[\overleftarrow{\mathbb{h}_{t+1}}, \overleftarrow{\mathbb{P}_{\text{Out}}}\right] + \overleftarrow{\mathbb{B}_{\text{Out}}}\right) \tag{11}$$

$$\overleftarrow{\mathbb{h}_t} = \overleftarrow{\text{Out}_{\text{Gate}}} * \tanh\left(\overleftarrow{C_t}\right) \tag{12}$$

where $\overleftarrow{C_t}$ is the cell state, $\overleftarrow{C_Inp_t}$ represents the candidate for $\overleftarrow{C_t}$ and $\overleftarrow{\mathbb{P}_{\text{For}}}$, $\overleftarrow{\mathbb{P}_{\text{Inp}}}$, $\overleftarrow{\mathbb{P}_c}$, and $\overleftarrow{\mathbb{P}_{\text{Out}}}$ denotes the inputs for the $\overleftarrow{\text{For}_{\text{Gate}}}$, $\overleftarrow{\text{Inp}_{\text{Gate}}}$, $\overleftarrow{C_Inp_t}$, and $\overleftarrow{\text{Out}_{\text{Gate}}}$, while $\overleftarrow{\mathbb{h}_{t+1}}$ represents the next LSTM block's output for the ($\leftarrow$) process. The $\overleftarrow{\mathbb{W}_{\text{For}}}$, $\overleftarrow{\mathbb{W}_{\text{Inp}}}$, $\overleftarrow{\mathbb{W}_c}$, and $\overleftarrow{\mathbb{W}_{\text{Out}}}$ are the weights. Further, $\overleftarrow{\mathbb{B}_{\text{For}}}$, $\overleftarrow{\mathbb{B}_{\text{Inp}}}$, $\overleftarrow{\mathbb{B}_c}$, and $\overleftarrow{\mathbb{B}_{\text{Out}}}$ are the Bias for the weights.

B. Self-Adaptive Attention Mechanism

In our approach, the self-adaptive attention mechanism dynamically learns to focus on the most relevant parts of the IS. Given a sequence of hidden states $\mathbb{M} = \{\mathbb{M}_1, \mathbb{M}_2, \ldots, \mathbb{M}_{\mathbb{Z}}\}$ from a BiLSTM layer, we compute the self-adaptive attention as follows. Each hidden state $\mathbb{M}_t$ is transformed into query $\mathbb{Q}_t$ and key $\mathbb{K}_t$ vectors through weight matrices $\mathbb{W}_q$ and $\mathbb{W}_k$, respectively, where t indexes the timestep within the sequence.

The self-adaptive attention weights $\mathbb{W}_t$ for each timestep t are computed based on the similarity between the query and key vectors across all timesteps, as shown below

$$\mathbb{S}_{t,j} = \mathbb{Q}_t^T \mathbb{K}_j \tag{13}$$

$$\mathbb{W}_t = \frac{\exp\left(\mathbb{S}_{t,t}\right)}{\sum_{j=1}^{\mathbb{Z}} \exp\left(\mathbb{S}_{t,j}\right)} \tag{14}$$

where $\mathbb{S}_{t,j}$ represents the attention score between the query vector at timestep t and the key vector at timestep j. The Softmax function is applied to ensure that the weights across all timesteps sum to 1, enabling a probabilistic interpretation of attention weights.

The output of the attention mechanism, $\mathbb{SV}$, is then computed as a weighted sum of the hidden states, scaled by the computed attention weights

$$\mathbb{SV} = \sum_{t=1}^{\mathbb{Z}} \mathbb{W}_t \mathbb{M}_t \tag{15}$$

where $\mathbb{SV}$ represents the context vector, capturing the most relevant information across the sequence as determined by the self-adaptive attention mechanism. This context vector is then used for subsequent processing or classification tasks, enabling the model to make informed decisions based on the dynamically learned importance of different parts of the IS.

C. XAI Mechanism

The DL-based IDS gained popularity in recent years due to its outstanding performance. However, they are treated as black boxes. XAI on the other hand explains and interprets the decision of IDS. It provides human users with rational and understandable justifications for their conduct [31]. Such an explanation and interpretation help the users to understand the rationale behind the IDS decision. In this work, we employed the SHAP mechanism of XAI to explain the decision of our proposed IDS. By figuring out how much each attribute contributed to the final judgment or prediction, SHAP explains

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Algorithm 1 Proposed Data-Efficient Algorithm for Intrusion Detection

Input: CICIDS2017 and X-IIoTID Datasets (D)
Normalize and preprocess D, extracting relevant features
Split D into training set D_{train} and testing set D_{test}
Function Train_Enhanced_BiLSTM_Model
Initialize weights $\mathbb{W}$ and biases $\mathbb{B}_{bias}$ for BiLSTM layers
Incorporate self-adaptive attention mechanism in the BiLSTM network
Define batch size $\mathbb{B}_{size}$ for training
for Epoch ← 1 to E **do**
    Batch training on D_{train} with size $\mathbb{B}_{size}$
    Apply self-adaptive attention to focus on relevant sequence parts
    Calculate and minimize the enhanced BiLSTM loss function
    Update $\mathbb{W}$, $\mathbb{B}_{bias}$, and attention parameters using back-propagation
end for
Return trained Enhanced BiLSTM model with self-adaptive attention
end Function
Function Test_Enhanced_BiLSTM_Model (model, D_{test})
while not End of D_{test} **do**
    Evaluate the model on D_{test}
    Generate and record output classifications
end while
Return test results
end Function
Enhanced_BiLSTM_Model ← Train_Enhanced_BiLSTM_Model()
Invoke Test_Enhanced_BiLSTM_Model with Enhanced_BiLSTM_Model and D_{test}
Assign output to Test_Results
Function Analyze_Results (Test_Results)
for each result in Test_Results **do**
    if result is 0 **then**
        Label as *Normal* activity
    else
        Label as one of the multiple *Attack* types (e.g., XSS, Botnet, etc.)
    end if
end for
Return analyzed results with multi-class attack identification
end Function
Final_Results ← Analyze_Results(Test_Results)
Output and save Final_Results

the predictions of an instance. The contribution can be positive or negative. The following equations can be used to obtain an instance's SHAP explanation:

$$\mathbb{G}(\mathbb{S}) = \mathbb{V}_o + \sum_{k=1}^{\mathbb{N}} \mathbb{V}_i \mathbb{S}_i \tag{16}$$

TABLE II
CICIDS2017 DATASET DETAILS

Class Type	Label	Instances
Benign	0	161465
FTP-Patator	1	7540
SSH-Patator	2	5449
DoS-GoldenEye	3	7938
Botnet	4	1960
PortScan	5	31480
DoS-Slowloris	6	5480
XSS	7	435

where $\mathbb{S}$ represents the simplified feature, $\mathbb{N}$ denotes the maximum size, and $\mathbb{V}_i$ represents the Shapely value SV. If the value of $\mathbb{V}_i$ is high for a feature, it means that this feature contributed the most to the decision of a model. Further, the most important features are selected by using the following equation:

$$\mathbb{IF}_j = \sum_{k=1}^{\mathbb{M}} ||\mathbb{V}_j(\mathbb{X}_i)|| \tag{17}$$

where $\mathbb{M}$ represents the number of samples, and $\mathbb{IF}_j$ denotes the average SV of the kth input feature.

D. Hyperparameters

We have investigated many parameters to find the most pre-eminent values to tune the proposed model. Several parameters, i.e., epochs, layers, neurons, and batch size were considered. We employed two layers of BiLSTM with 100 and 50 neurons followed by one dense layer with 20 neurons. To avoid overfitting, we used a dropout of 0.2%. We kept the number of layers to two because the results stop to improve after two layers. Integrated after the BiLSTM layers, the self-adaptive attention mechanism assigns different weights to various inputs in the sequence, enabling the model to focus on the most relevant features for accurate intrusion detection. Further, we performed the simulations for ten epochs with a batch size of 64. Moreover, we used ADAM optimizer for optimization, RELU as the activation function, and CC-E as the loss function. Finally, the Softmax classifier is used in the output layer. The schematic architecture of the proposed IDS is presented in Fig. 1.

IV. EXPERIMENTAL SETUP AND EVALUATION METRICS

This section presents the complete details of the experimental setup, datasets, and preprocessing. We further provide details about the metrics we used for the evaluation of the proposed threat detection model.

A. Experimental Setup

The experimentation was performed using an Alienware PC, processing @ 2.7 GHz with 32-GB RAM and an RTX 4080 GPU. Further, we employed the Keras library of Tensorflow to develop the proposed IDS and used Python to run the scripts.

1) Dataset and Preprocessing: In this work, we employed the CICIDS2017 [32] and X-IIoTID [33] datasets to evaluate the performance of the proposed IDS. These datasets contain

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TABLE III
X-IIoTID DATASET DETAILS

Class Type	Label	Instances
Normal	0	421417
Reconnaissance	1	127590
RDoS	2	141261
Weaponization	3	67260
C&C	4	2863
crypto-ransomware	5	154
Exfiltration	6	5094
Exploitation	7	1133
Lateral_movement	8	31596
Tampering	9	5094

real-time IoT network traffic of Normal and attack instances. The attack classes include RDoS, XSS, DoS GoldenEye, SSH, FTP Patator, Botnet, PortScan C&C, Exploitation, Tampering, and many more. However, this work concerns eight classes of the CICIDS2017 dataset, including seven classes of attacks and one normal class, and ten classes of X-IIoTID, including nine classes of attacks and one normal class. The complete details about the classes and instances used for experiments are provided in Tables II and III. Further, the dataset is divided into two sets, i.e., the Training set and the Testing set [34]. We used 70% data from the CICIDS2017 and X-IIoTID datasets to train the proposed IDS and the remaining 30% to test and validate its performance. Finally, we employed the necessary steps to preprocess and normalize the datasets [30].

B. Evaluation Metrics

Various metrics of evaluation have been employed to thoroughly evaluate the performance of the proposed IDS, i.e., Accuracy ($\mathbb{Acc}$), Precision ($\mathbb{Pre}$), Recall ($\mathbb{Rec}$), F1-score ($\mathbb{F1}$), Receiver Operating Characteristics ($\mathbb{Roc}$) Curve, Confusion Matrix ($\mathbb{Cm}$), etc. The following equations are used to calculate the $\mathbb{Acc}$, $\mathbb{Pre}$, $\mathbb{Rec}$, and $\mathbb{F1}$:

$$\mathbb{Acc} = \frac{\mathbb{Tpr} + \mathbb{Tnr}}{\mathbb{Tpr} + \mathbb{Tnr} + \mathbb{Fpr} + \mathbb{Fnr}} \tag{18}$$

$$\mathbb{Pre} = \frac{\mathbb{Tpr}}{\mathbb{Tpr} + \mathbb{Fpr}} \tag{19}$$

$$\mathbb{Rec} = \frac{\mathbb{Tpr}}{\mathbb{Tpr} + \mathbb{Fnr}} \tag{20}$$

$$\mathbb{F1} = 2 * \frac{\mathbb{Pre} * \mathbb{Rec}}{\mathbb{Pre} + \mathbb{Rec}} \tag{21}$$

where $\mathbb{Tpr}$ and $\mathbb{Tnr}$ are true positive and true negative rates, while $\mathbb{Fpr}$ and $\mathbb{Fnr}$ represents the false positive and false negative rates.

V. RESULTS AND DISCUSSION

In this section, we discuss and analyze the performance of the proposed IDS, followed by its interpretation. Finally, Finally, we will compare it with some recent IDSs from the current literature.

The degree to which a model accurately recognizes or forecasts outcomes from the dataset it was trained on is known as its training $\mathbb{Acc}$. The model's ability to identify patterns and correlations in the training data is demonstrated

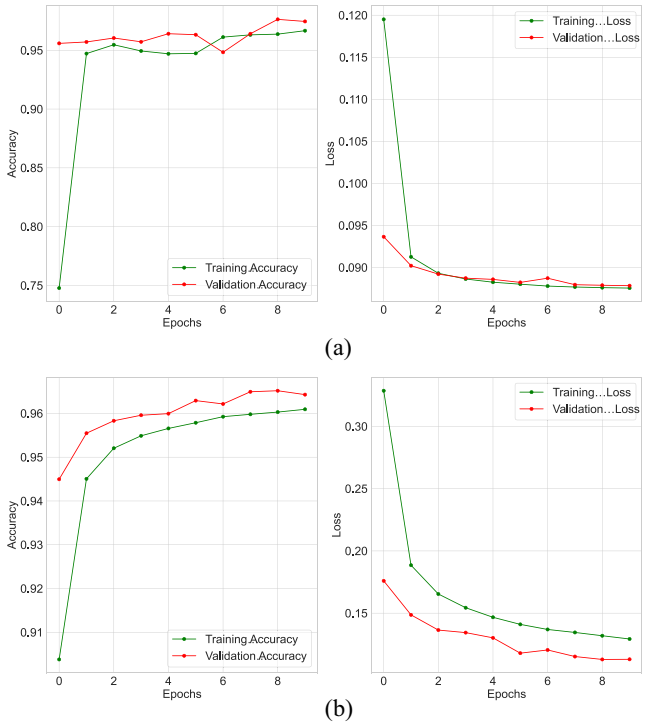


Fig. 2. $\mathbb{Acc}$ versus loss of the proposed IDS. (a) CICIDS2017 dataset. (b) X-IIoTID dataset.

by high-training $\mathbb{Acc}$. However, assessing a model's overall performance requires more than just one statistic. Validation $\mathbb{Acc}$ is important in this situation. Validation $\mathbb{Acc}$ evaluates how well the model performs on an independent, unseen dataset. It offers perceptions of the model's capacity to generalize its learning to new data, which is crucial to its usefulness in real-world scenarios. Overfitting is a prevalent problem that arises when a model performs well during training but badly during validation. Therefore, we present the testing $\mathbb{Acc}$ versus validation $\mathbb{Acc}$ of the proposed threat detection framework in Fig. 2. Fig. 2(a) depicts the testing versus validation $\mathbb{Acc}$ using the CICIDS2017 dataset, while Fig. 2(b) presents it for the X-IIoTID dataset. It can be seen that the model has shown an efficient performance in terms of validation $\mathbb{Acc}$ and proves that the model is not overfitting.

Furthermore, loss functions offer a more nuanced perspective by measuring the difference between expected and actual results. Training loss measures how well the model performs on the training data, with lower values indicating better performance. However, the error rate on the unseen validation dataset is quantified as validation loss. Ensuring that training and validation loss reduce over time and at a similar pace is a key challenge in model development. Overfitting is indicated by a divergence, where training loss keeps going down but validation loss goes up. This indicates that the model's prediction ability on unseen, validation data is declining due to its overfitting to the training set. Fig. 2(a) depicts the training versus validation loss using the CICIDS2017 dataset, while Fig. 2(b) depicts the training versus validation loss using the X-IIoTID dataset. It can be seen that both the training and validation loss decrease with time, thus proving that the

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TABLE IV
CLASSWISE PERFORMANCE OF THE PROPOSED IDS USING CICIDS2017

Metrics	Benign	FTP-Patator	SSH-Patator	DoS-GoldenEye	Botnet	PortScan	DoS-Slowloris	XSS
Precision ($\mathbb{Pre}$)	99.89	99.60	99.68	96.28	98.54	99.91	99.71	99.67
F1-score ($\mathbb{F1}$)	99.88	99.60	99.78	97.64	98.51	99.69	99.64	98.41
Recall ($\mathbb{Rec}$)	99.87	99.60	99.88	99.04	98.48	99.47	99.21	97.17
False Positive Rate ($\mathbb{Fpr}$)	0.72265	0.00	0.43167	0.31943	0.00	0.00	0.89992	0.10426

TABLE V
CLASSWISE PERFORMANCE OF THE PROPOSED IDS USING X-IIOTID

Metrics	Normal	Reconnaissance	RDoS	Weaponization	C&C	crypto-ransomware	Exfiltration	Exploitation	Lateral_movement	Tampering
Precision ($\mathbb{Pre}$)	95.67	96.91	99.92	95.19	99.76	98.33	91.70	98.06	97.87	99.63
F1-score ($\mathbb{F1}$)	97.18	93.47	99.92	96.86	66.71	95.93	92.09	72.89	91.24	82.71
Recall ($\mathbb{Rec}$)	98.74	90.26	99.92	98.59	50.11	93.65	92.48	58.00	85.45	70.70
False Positive Rate ($\mathbb{Fpr}$)	0.97183	0.93471	0.99925	0.96865	0.66719	0.95934	0.92096	0.72890	0.91244	0.82711

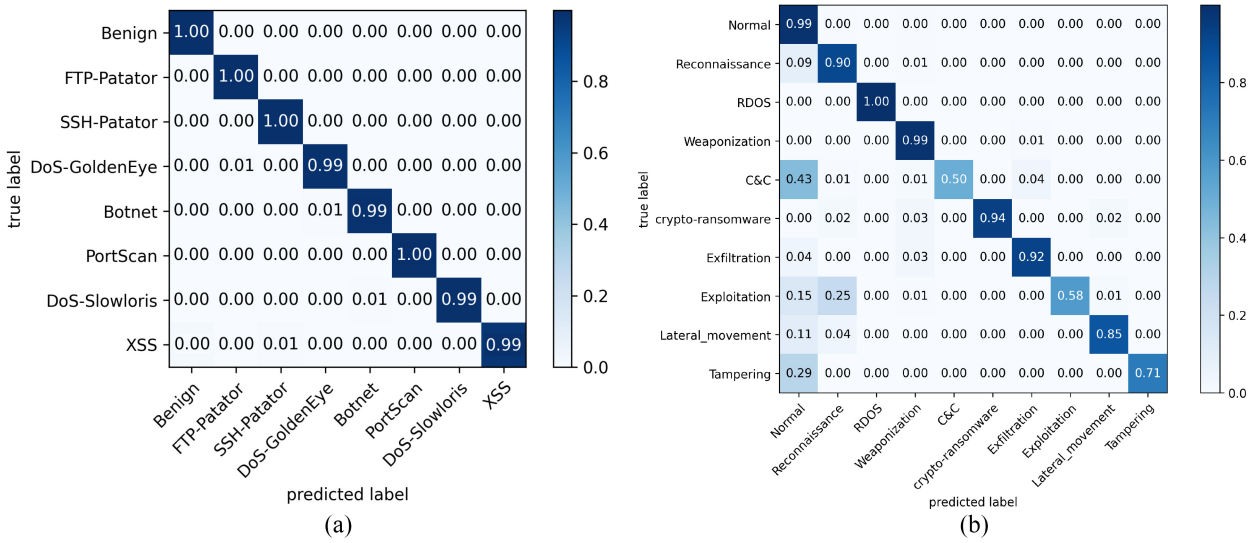


Fig. 3. $\mathbb{Cm}$ of the proposed IDS. (a) CICIDS2017 dataset. (b) X-IIoTID dataset.

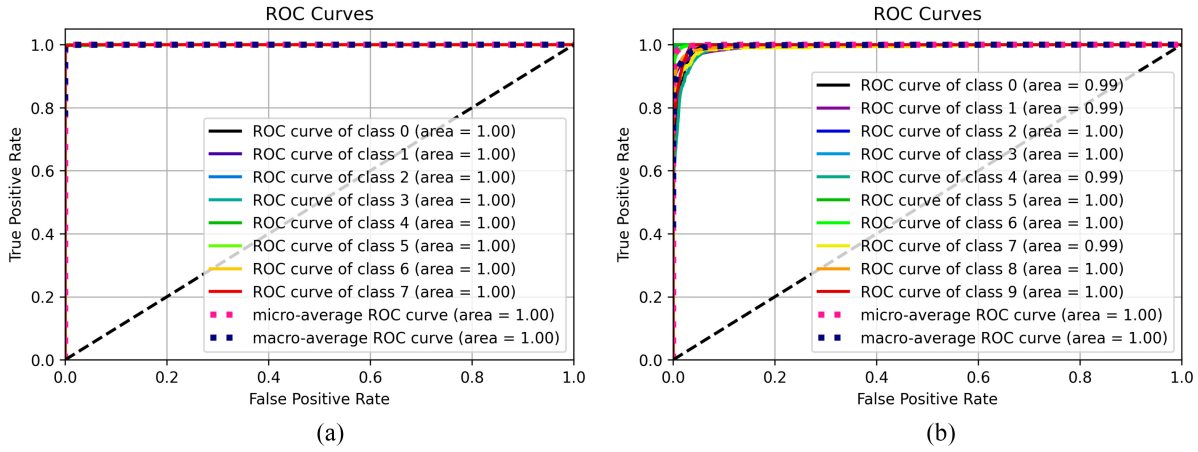


Fig. 4. $\mathbb{Roc}$ of the proposed IDS. (a) CICIDS2017 dataset. (b) X-IIoTID dataset.

proposed model is not overfitting and is an optimal fit. A $\mathbb{Cm}$ is used in ML and DL-based threat detection schemes to show how well a classification model is performing. It displays the number of accurate and inaccurate predictions the model made on test data. It is typically depicted as a square matrix, with the actual and predicted classes represented by the rows and columns, respectively. Fig. 3(a) depicts the $\mathbb{Cm}$ of our proposed scheme using the CICIDS2017 dataset, while Fig. 3(b) depicts the $\mathbb{Cm}$ using the X-IIoTID dataset. It can be seen that it identified all of the classes of the dataset efficiently with a high $\mathbb{Tpr}$ and low $\mathbb{Fpr}$. Additionally, The $\mathbb{Roc}$ offers valuable intuits into the tradeoff between the $\mathbb{Tpr}$ and the $\mathbb{Fpr}$. The performance of the classification model is frequently assessed using the Area Under the Curve ($\mathbb{AUC}-\mathbb{ROC}$) as

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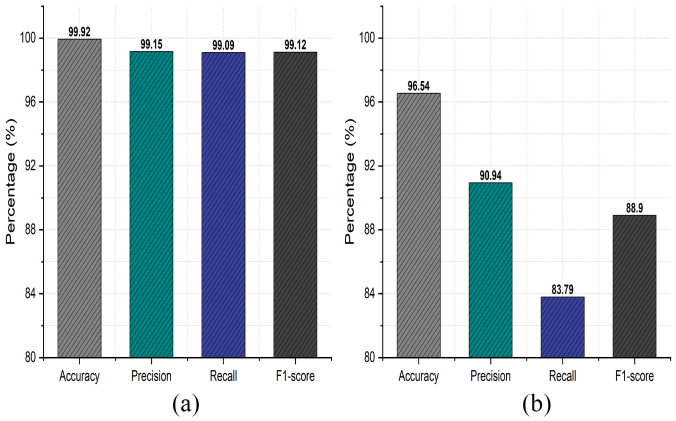


Fig. 5. Overall performance analysis of the proposed IDS. (a) CICIDS2017 dataset. (b) X-IIoTID dataset.

a summary metric. Better discrimination power and model performance are indicated by a larger $\mathbb{AUC} - \mathbb{Roc}$ value; a perfect classifier has a $\mathbb{AUC} - \mathbb{Roc}$ of 1. The $\mathbb{Roc}$ of our proposed IDS is presented in Fig. 4(a) for the CICIDS2017 and Fig. 4(b) for the X-IIoTID dataset. The effectiveness of the proposed IDS is demonstrated by the fact that the $\mathbb{ROC}$ curve for the proposed scheme is 1 for both datasets.

Moreover, an IDS can be considered efficient if it achieves high values for $\mathbb{Pre}$, $\mathbb{F1}$, $\mathbb{Rec}$, and low values for $\mathbb{Fpr}$. We provide the classwise performance of our proposed IDS in Table IV using the CICIDS2017 dataset. It has achieved the highest $\mathbb{Pre}$ of 99.91% and the lowest of 96.28%. Regarding $\mathbb{F1}$, it has achieved values between 97.64% to 99.88%. For $\mathbb{Rec}$, it has the highest $\mathbb{Rec}$ of 99.88% for the SSH-Patator class and the lowest of 97.17% for the XSS class. Finally, it has achieved the lowest $\mathbb{Fpr}$ of 0.00% for FTP-Patator, Botnet, and PortScan classes and the highest of 0.8992% for the XSS class. The classwise performance using the X-IIoTIDT dataset is provided in Table V. The proposed IDS achieved $\mathbb{Pre}$ between 91.70% to 99.92% for all the classes. Regarding $\mathbb{F1}$ and $\mathbb{Rec}$, it has achieved the highest $\mathbb{F1}$ of 99.92% RDOS class and lowest of 66.71% for C&C class. Regarding other classes, it has $\mathbb{F1}$ between 72.89% to 96.86%, respectively. Further, it has the highest $\mathbb{Rec}$ of 99.92% for RDOS class and lowest of 50.00% and 58.00% for C&C and Exploitation classes. For other classes, it achieved $\mathbb{Rec}$ values in between 70.70% and 98.74%. Moving forward, it has the highest $\mathbb{Fpr}$ of 0.99925 for the RDoS class and the lowest $\mathbb{Fpr}$ of 0.82711% for the tampering class. Furthermore, we provide the overall performance analysis of our proposed IDS in terms $\mathbb{Acc}$, $\mathbb{Pre}$, $\mathbb{Rec}$, and $\mathbb{F1}$. Fig. 5(a) depicts the performance analysis using the CICIDS2017 dataset. The proposed framework achieved $\mathbb{Acc}$ OF 99.92% with $\mathbb{Pre}$ of 99.15%, $\mathbb{Rec}$ of 99.09%, and $\mathbb{F1}$ of 99.22%. Regarding, the X-IIoTID dataset, it achieved 96.54% $\mathbb{Acc}$, 90.94 $\mathbb{Pre}$ and $\mathbb{Rec}$ and $\mathbb{F1}$ of 83.79% and 88.90%, respectively.

Moreover, we used the Summary Plot ($\mathbb{S}_{\text{plot}}$), Waterfall Plot ($\mathbb{W}_{\text{plot}}$), and Decision Plot ($\mathbb{D}_{\text{plot}}$) of the SHAP mechanism of XAI to interpret the predictions of our proposed IDS. The $\mathbb{S}_{\text{plot}}$ presents the essential features that contribute the most to the model decision-making. It presents the important

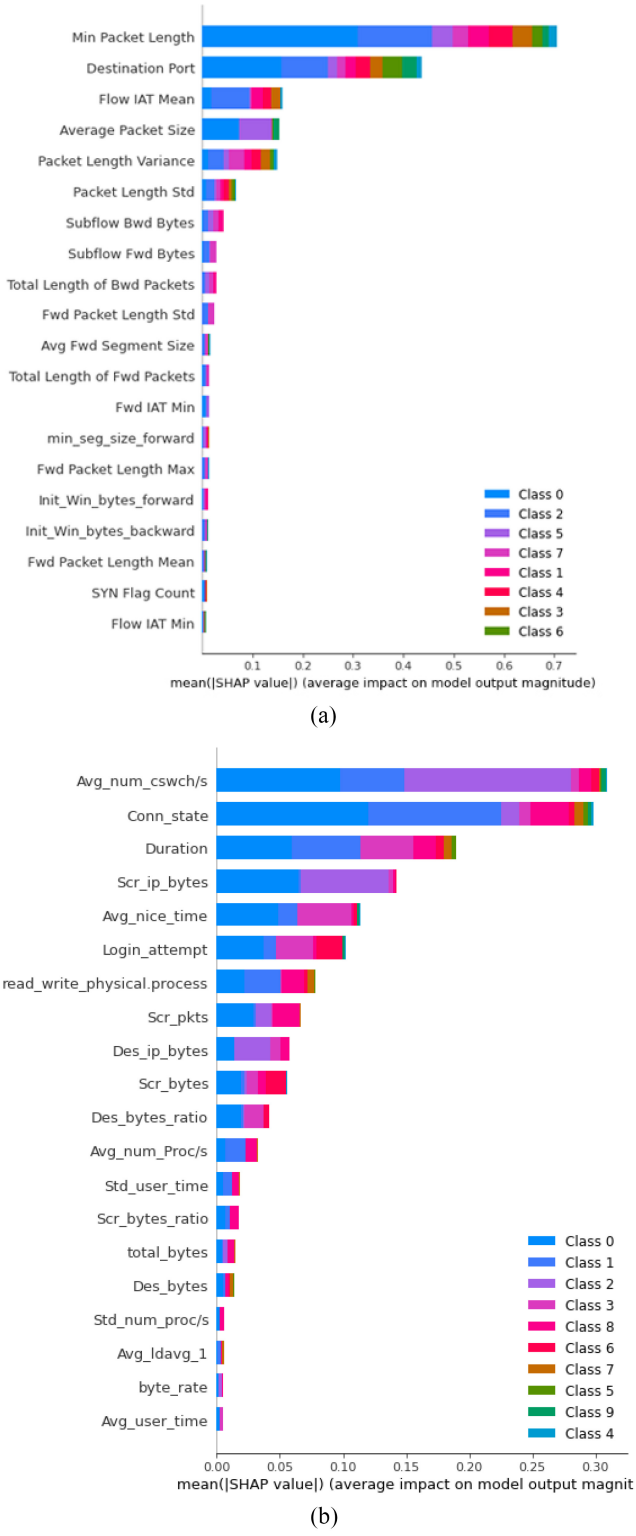


Fig. 6. SHAP values using $\mathbb{S}_{\text{Plot}}$. (a) CICIDS2017 dataset. (b) X-IIoTID dataset.

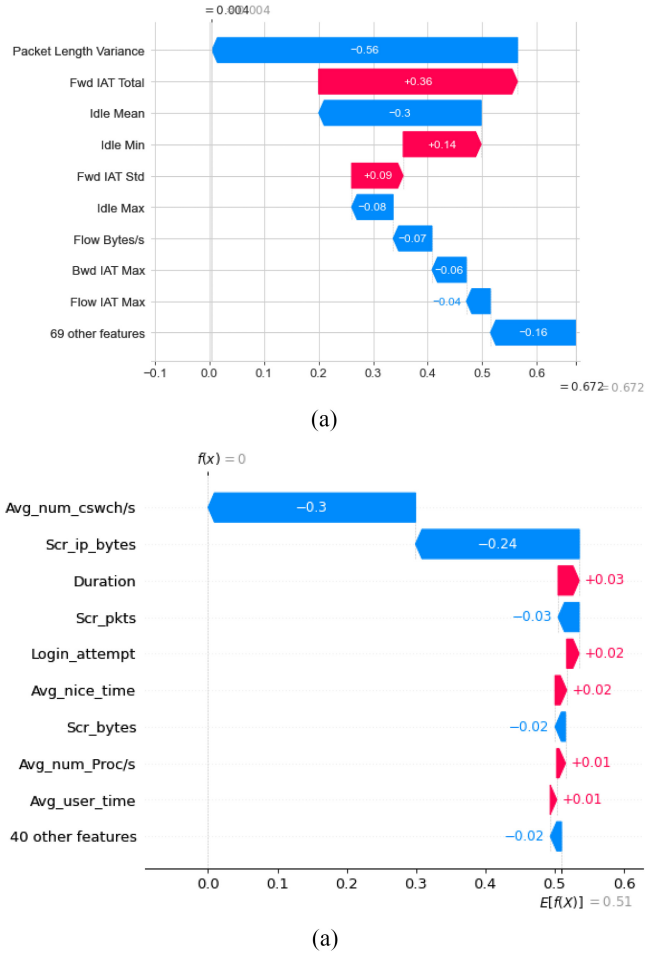
features at the top, while the less important ones are at the bottom. Such a plot greatly impacts the model decision, so the features with high magnitude are considered the most significant. Fig. 6 depicts the $\mathbb{S}_{\text{plot}}$, where Fig. 6(a) depicts the $\mathbb{S}_{\text{plot}}$ using the CICIDS2017 and Fig. 6(b) depicts the $\mathbb{S}_{\text{plot}}$ using the X-IIoTID datasets. These figures present the most

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TABLE VI
COMPARISON WITH EXISTING WORKS

Authors	Year	Model	XAI	Dataset	Acc	Pre	Rec	F1
[35]	2018	CNN	×	CICIDS2017	82.00%	-	-	89.00%
[26]	2021	BiLSTM	×	KDD-CUP99, UNSW-NB15	99.50%	-	-	-
[9]	2022	GT-CSDNN	×	CICIDS2017	97.96%	98.65%	98.47%	99.00%
Proposed IDS	2023	SABiLSTM	✓	CICIDS2017	99.92%	99.15%	99.09%	99.12%
				X-IIoTID	96.54%	90.94%	83.79%	88.90%

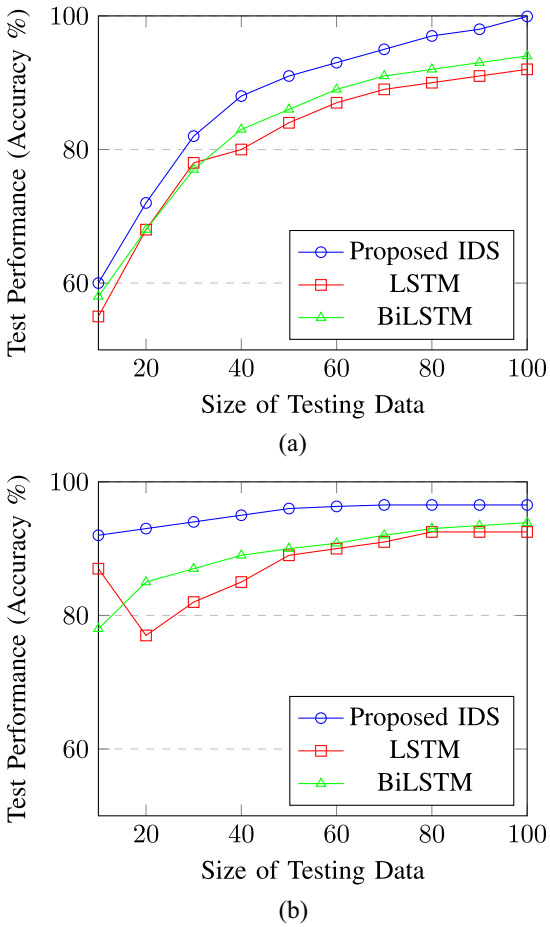
Terms & Abbreviations: CNN: Convolutional Neural Network; MLP: Multilayer Perception; DNN: Deep Neural Network; SABiLSTM: self-adaptive attention Bidirectional Long short-term Memory Network.



(a)

(a)

Fig. 7. SHAP values using $\mathbb{W}_{\text{Plot}}$. (a) CICIDS2017 dataset. (b) X-IIoTID dataset.



(a)

(b)

Fig. 8. Comparison of data-efficiency with some baseline models. (a) Learning curve on testing data for CICIDS2017 dataset. (b) Learning curve on testing data for X-IIoTID dataset.

and least important features of the datasets, contributing to our proposed IDS's decision-making. Additionally, a $\mathbb{W}_{\text{plot}}$ illustrates the progressive addition or subtraction of the impact of each feature from an initial baseline or baseline prediction to the forecast produced by the model. It enables the prediction to be broken down into the various contributions of the input variables. It starts with the original forecast or starting point. The impact of each attribute on prediction is then demonstrated as shown in Fig. 7(a) for the CICIDS2017 dataset and Fig. 7(b) for the X-IIoTID dataset. It further shows the overall effect of all the features up to that point when each feature's contribution is increased or decreased. The red and blue bars present the features that contribute the most to the classification score of the proposed model. Finally, it provides the final prediction, which is the cumulative contributions of all features.

A. Comparison of Data Efficiency With Some Commonly Used Models

Fig. 8 illustrates the learning curves of three models—proposed IDS, LSTM, and BiLSTM—over two different datasets, offering insights into the data efficiency of these models, particularly focusing on the performance of the proposed IDS. Fig. 8(a) showcases the performance on CICIDS2017 Dataset. Here, the proposed IDS model outperforms the LSTM and BiLSTM models, achieving higher accuracy with less training data. This characteristic is a hallmark of data efficiency, indicating that the proposed IDS model requires

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fewer data points to learn and adapt, making it more suitable for scenarios with limited data availability. The steeper slope of the proposed IDS curve, especially in the initial stages, suggests that it is capable of extracting more meaningful information from smaller datasets compared to the other models. Fig. 8(b) depicts the learning curve on the X-IIoTID dataset, where the proposed IDS model starts with a notably high accuracy of 92% even with the smallest training dataset size. This again reflects the model's superior data efficiency, as it is able to achieve high performance without the need for large amounts of data. The ability of the proposed IDS model to maintain its lead in performance with increasing data sizes further solidifies its position as a data-efficient model.

B. Comparison With Recent IDSs From Literature

At last, we compare the proposed IDS performance with recent IDSs from the current literature to further prove its superiority, i.e., [9], [26], and [35]. Table VI depicts the complete comparison. Kim et al. [35] employed a CNN-based approach for developing a new encoding mechanism to identify anomalous activities in the network. They trained their proposed IDS on the CICIDS2017 dataset and achieved $\mathbb{Acc}$ of 82% and $\mathbb{F}1$ of 89%, respectively. They haven't provided any details about the $\mathbb{Pre}$ and $\mathbb{Rec}$. Further, Pooja and Shrinivasacharya [26] employed a BiLSTM-based approach for threat detection. They trained their proposed IDS on the KDD-CUP99 and UNSW-NB15 datasets and achieved an average $\mathbb{Acc}$ of 99.50%, respectively. The results in terms of $\mathbb{Pre}$, $\mathbb{Rec}$, and $\mathbb{F}1$ is missing in their article. Moreover, Balamurugan et al. [9] have designed an efficient IDS based on game theory and DNN (GT-DNN) to improve the security of cloud computing. They have trained their model using the CICIDS2017 dataset and attained $\mathbb{Acc}$ of 97.96%, $\mathbb{Pre}$ of 98.65%, $\mathbb{Rec}$ of 98.47%, and $\mathbb{F}1$ of 99%. The comparison in Table VI shows that our proposed framework outperformed the recent frameworks, thus further proving its efficacy.

VI. Conclusion

This article developed an enhanced and explainable threat detection system for IIoT networks by combining explainable artificial intelligence with a deep learning-based methodology. In particular, We use BiLSTM with a self-adaptive attention mechanism and Softmax to develop an IDS. Additionally, we utilized the SHAP mechanism of explainable artificial intelligence to analyze the key features of the datasets that influence the model's decision-making to clarify the prediction or decision of our proposed IDS. This explanation of the decision will make it easier for security analysts to comprehend the reasoning behind our proposed IDS decision. The proposed IDS achieved a high-detection accuracy of 99.92\% and 96.54\%, respectively. The performance of the proposed IDS is compared with traditional and recent frameworks from the current literature. Upon comparison, the proposed IDS outclassed the other schemes. Thus, endorses the dominance of the proposed IDS against these frameworks and proves it an efficient security solution for such an industrial network. Future research will combine blockchain and FL to design an IDS for the Industrial network.

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